

AI Chatbot for Instant Home Service Booking

1. Introduction

This project proposes an AI-powered multilingual chatbot that enables users to book home services instantly using natural language in English, Hindi, or Hinglish. The system analyze user input, detects service category, urgency level, and location, and automatically converts the conversation into structured booking data.

The chatbot simplifies the booking process and ensures service assistance within 15 minutes using smart automation.

2. System Solution Overview

The proposed system consists of the following modules:

1. Chat Interface
2. Natural Language Processing (NLP) Engine
3. Intent & Entity Detection Module
4. Booking Data Formatter
5. Booking API Integration
6. Confirmation System
7. Chat History Management

3. Working of the System

Step 1: User Interaction

User sends message in English, Hindi, or Hinglish.

Example:

“Mujhe kitchen clean karwana hai urgent.”

Step 2: Language Detection

System detects the language automatically.

Step 3: Intent Detection

System identifies whether the user wants:

- New Booking

- Cancellation
- Inquiry
- Reschedule

Step 4: Entity Extraction

The chatbot extracts:

- Service Type → Deep Cleaning
- Area → Kitchen
- Priority → Urgent
- Location → (if mentioned)

Step 5: Structured Booking Conversion

Example JSON Format:

```
{  
Service: Deep Cleaning  
Area: Kitchen  
Priority: High  
Location: Gurgaon  
Time: ASAP  
}
```

Step 6: API Integration

The structured data is sent to the backend booking API, which:

- Finds nearest available service provider
- Confirms availability
- Assigns service partner

Step 7: Instant Confirmation

User receives confirmation message:

Booking Confirmed
Service: Kitchen Deep Cleaning
Arrival Time: 12 Minutes
Booking ID: BK10245

4. Key Features

- Understands English, Hindi, and Hinglish
- Automatically detects service category
- Identifies urgency level from text
- Converts conversation into structured booking data
- Suggests correct service if input is unclear
- Integrates with booking system API
- Provides instant confirmation
- Maintains chat history

5. Tools and Technologies Required

1. Frontend Development Tools

- React.js (Web Interface)
- Flutter (Mobile App)
- HTML, CSS, JavaScript
- WhatsApp API (Optional integration)

2. Backend Development Tools

- Node.js / Express.js
OR
- Python (FastAPI / Django)

3. AI & NLP Tools

- OpenAI API (for Natural Language Understanding)
- Rasa (Conversational AI Framework)
- Hugging Face Transformers
- Google ML Kit (Language Detection)

4. Database

- MongoDB (NoSQL Database)
OR
- PostgreSQL (Relational Database)

5. Cloud & Deployment

- AWS / Google Cloud / Azure
- Firebase (Real-time database & hosting)
- Docker (Containerization)

6. APIs Required

- Booking Management API
- Location Services API (Google Maps API)
- Notification API (SMS / Email)
- Payment Gateway API (Optional)

6. System Architecture

User → Chat Interface → NLP Engine → Intent & Entity Extraction → Structured Booking Generator → Booking API → Confirmation → User

7. Advantages of the System

- Faster booking process
- Reduced manual errors
- Supports regional language users
- Improves customer experience
- Automation reduces operational cost

8. Conclusion

The AI-powered chatbot simplifies the home service booking process by understanding natural language input and converting it into structured booking data

instantly. It ensures quick service assignment within 15 minutes and enhances user experience through intelligent automation.